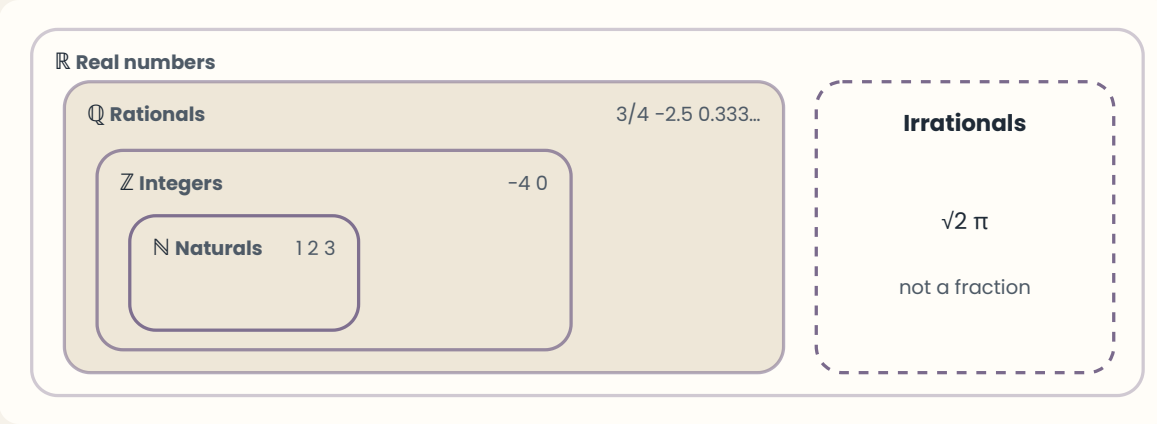
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Calculus 1 · Week 1

Introduction and inequalities

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, and the irrationals complete $\mathbb{R}$.



> **Think of it like this.** Think of the number line as a long street where every address exists, even 2.5 and √2. "Less than" just means "further west".

Move everything to one side: $P(x) = 0$.

- The degree is the highest power after simplifying.
- Degree n means at most n real roots; check roots by substitution.

> **Think of it like this.** An equation is a balance scale. Whatever you do to one pan, you do to the other, and the scale stays level.

Linear: one root, x = −b/a.

- Quadratic: factor, isolate x^2, or use the formula.
- $\Delta = b^2 - 4ac$ decides 2, 1, or 0 real roots; in word problems discard roots that make no sense.

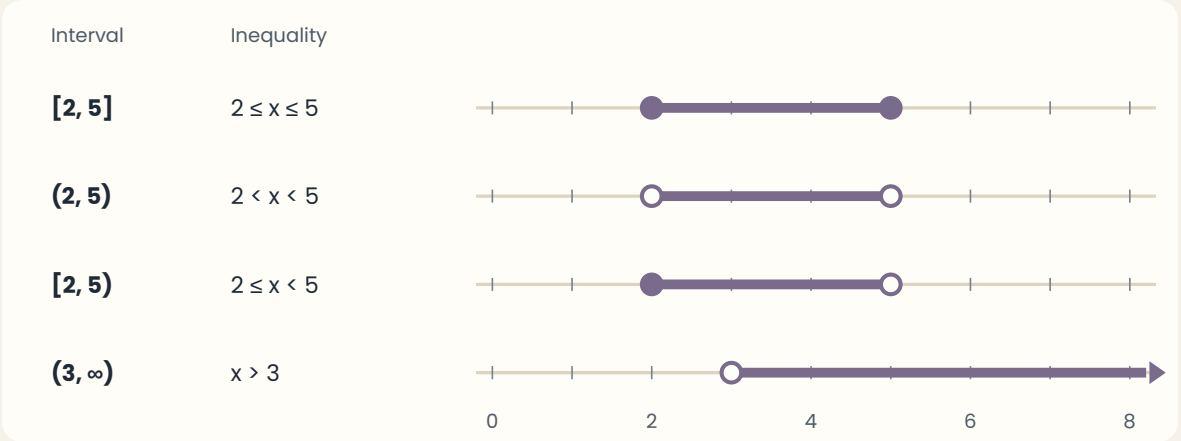
> **Think of it like this.** The discriminant is a weather forecast for roots: you learn whether to expect two, one, or none before you go out and compute.

Rational root candidates: ± (divisors of the constant) / (divisors of the leading coefficient).

- Remainder 0 in synthetic division means you found a root.
- After squaring or clearing denominators, check every answer.

> **Think of it like this.** Synthetic division is like peeling an onion: each root you find removes one layer, until the core is a quadratic.

[ ] includes the endpoint, ( ) excludes it; ∞ always takes ( ).



> **Think of it like this.** A bracket is a closed door you can stand in; a parenthesis is a door you can walk up to but never step through.

Add, subtract, or multiply by a positive: the sign stays.

- Multiply or divide by a negative: flip the sign.
- Check one value from your answer in the original inequality.

> **Think of it like this.** Multiplying by a negative number is like looking in a mirror: left and right swap, so "smaller" becomes "larger".

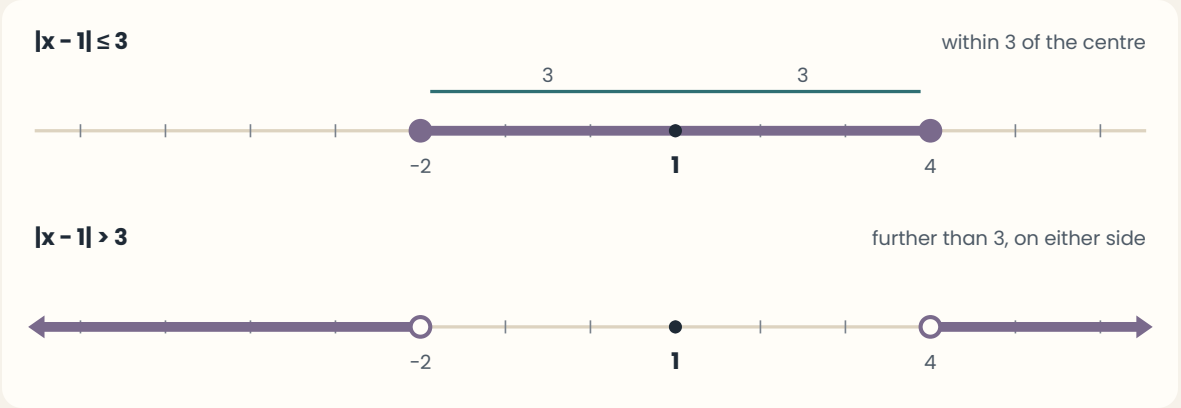
Zero on one side, factor, find critical points, test one value per interval.

Interval	$x < -4$	$-4 < x < 3$	$x > 3$
Test value	$x = -5$	$x = 0$	$x = 4$
$x + 4$	−	+	+
$x - 3$	−	−	+
$(x + 4)(x - 3)$	+	−	+



> **Think of it like this.** Critical points are the traffic lights of the number line. Between two lights the colour cannot change, so you only need to look once per block.

$|x - a| < r \rightarrow a - r < x < a + r$ (one interval).



Think of it like this. |x − a| < r is a dog on a leash of length r tied at a: it can be anywhere within r of the post, on either side.

Step by step

Steps of a sign chart

1. Move everything to one side: $x^2 + 2x - 3 > 0$
2. Factor: $(x + 3)(x - 1) > 0$
3. Mark the critical points -3 and 1
4. Test -4, 0, and 2 in the factors
5. Keep the positive intervals: $(-\infty, -3) \cup (1, \infty)$

Check yourself

1. Which set contains $\sqrt{7}$?

A) The integers $\mathbb{Z}$ B) The rationals $\mathbb{Q}$ C) The reals $\mathbb{R}$

2. What is the degree of $x(x - 1) = x^2 + 4$?

A) 2 B) 1 C) 0

3. How many real roots has $4x^2 - 4x + 1 = 0$?

A) Two B) One (repeated) C) None

Answers are in the speaker notes.

Practise next

Labs

- Equations warm-up
- Linear inequalities and intervals
- Sign charts
- Absolute value, systems, and a model

On the site

- Inequality Lab